

ISYE 6202 Supply Chain Facilities ISYE 6335 Supply Chain Engineering 1

Casework 1
Tsukumo Supply Chain Facility Network
Analysis, Planning, and Simulation

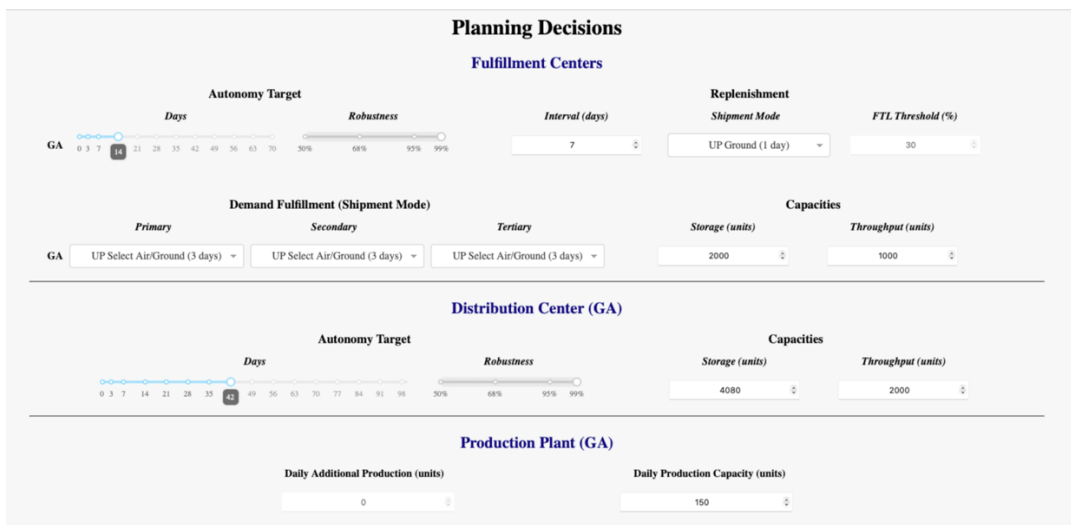
Professor Benoit Montreuil and Doctoral Student Ashwin S. Pothen

- *The casework is meant to be realized in self-organized teams of up to 6 students, yet may be realized solo, with the number of students in the team having no influence on the grading.*
- *The casework is shaped around two equally weighted (50%) main phases, each with a set of mandatory tasks having a significant grading weight, purposefully not revealed to the class.*
- *The casework must be completed and the report delivered by October 8th.*
- *Phase 1 must be realized collaboratively within each team. In Phase 2, each student must realize tasks individually in a coopetition mode with teammates, competing for best performance yet allowing discussions between teammates to stimulate collective learning.*
- *Grading will be team-based for Phase 1 deliverables. In Phase 2, grading will be individual, reflecting each student's independent work, while still encouraging team consultation as the reports must account for relative positioning of the results of all teammates.*
- *For each task, you must rigorously describe your methodology, present your results, analyze them, and discuss them relative to the current task and, when pertinent, relative to previous tasks.*
- *For computational algorithms and simulations, exploitation of Python and linked tools is preferred. Using spreadsheets such as Microsoft Excel/VBA is tolerated. All developed codes must be provided. For maps and visualizations, exploiting Python libraries or Tableau (or similar analytics software) is recommended.*
- *Phase 1 must be completed at the very latest by September 24th at 23h59, with the report including all its tasks 1-10 deliverables submitted online.*
- *Upon completion of phase 1, at any time before or on September 24th, each team will be provided with a simulator and a user guide to be leveraged for realizing the tasks of phase 2.*
- *In phase 2, the simulator-based team experience comprises nine levels, with successive level access and documentation released once the team completes the previous level. Phase 2 must be completed by October 8th at 23h59 with all simulation levels completed and all task deliverables submitted online.*
- *Starting in phase 1, then within the levels of phase 2, it is strongly encouraged to read the available casework documentation to identify early the types of tasks to be realized and the potential for proactive concurrent preparation. This is crucial for teams to plan and not simply react, to leverage the potential and availability of each team member, to avoid a lengthy linear single-thread realization of the casework tasks, and to rather achieve concurrent-engineering-style task realization.*
- *The answers must be limited to a maximum of 3 pages per task.*

Introduction to the Simulator Used in this Casework

Imagine stepping into the role of a chief supply chain officer (CSCO) at an integrated multinational manufacturer-retailer giant, managing an intricate supply chain system. Your team's responsibilities span from daily production planning at the factory to optimizing inventory levels across a multi-echelon warehouse network, all towards maximizing customer satisfaction and your firm's profitability. As a hands-on leader, you would interact regularly with key decision-makers and be required to have a broad vision and understanding of the impact of various strategic, tactical, and operational decisions. On a more detailed level, you need to make decisions about demand planning, production planning, inventory management, transportation and shipment management. While these tasks may seem daunting, a sophisticated simulator will give you the tools to operate like a true supply chain leader.

This Simulator, developed by Georgia Tech's Physical Internet Center (led by Prof. Montreuil), allows you to effortlessly set key parameters of a complex supply chain system, such as inventory targets, delivery modes, and production capacities, through an intuitive set of interactive tools (Figure 1). It then simulates daily demand to evaluate the effectiveness of your current supply chain system. The simulator vividly displays all order statuses and inventory levels on an interactive map, while also tracking your revenue and costs. Additionally, it offers a range of other exciting features and visualizations for you to explore! What would typically take a team of analysts weeks to evaluate can now be simulated in just minutes.



Planning Decisions

Fulfillment Centers

Autonomy Target: Days (0 to 70), Robustness (30% to 99%)

Interval (days): 7

Replenishment Shipment Mode: UP Ground (1 day)

FTL Threshold (%): 30

Demand Fulfillment (Shipment Mode)

Primary: UP Select Air/Ground (3 days)

Secondary: UP Select Air/Ground (3 days)

Tertiary: UP Select Air/Ground (3 days)

Capacities

Storage (units): 2000

Throughput (units): 1000

Distribution Center (GA)

Autonomy Target: Days (0 to 35), Robustness (30% to 99%)

Storage (units): 4080

Throughput (units): 2000

Production Plant (GA)

Daily Additional Production (units): 0

Daily Production Capacity (units): 150

Figure 1: Interactive interface for setting various decision parameters in the simulator

To give you an idea of what you'll experience:

- Dynamic Demand Simulation and Forecasting:** The engine simulates customer demand from various locations in the USA, and dynamically utilizes the fulfillment network to satisfy the demand. It also refreshes demand forecasts on a daily basis, based on the historical information up to the previous day, utilizing a sophisticated algorithm considering the various seasonalities and trends.

- Dynamic Inventory Planning:** With the daily-updated demand forecasts, the simulator dynamically adjusts production plans and inventory levels at each node of the network by just specifying your inventory targets (say you want to maintain a robust autonomy of 4 weeks of stock in the network with 99% confidence that no demand is lost during these four weeks). It enables for example to visualize the evolving current state of the inventory levels and flow of goods through the network (Figure 2).

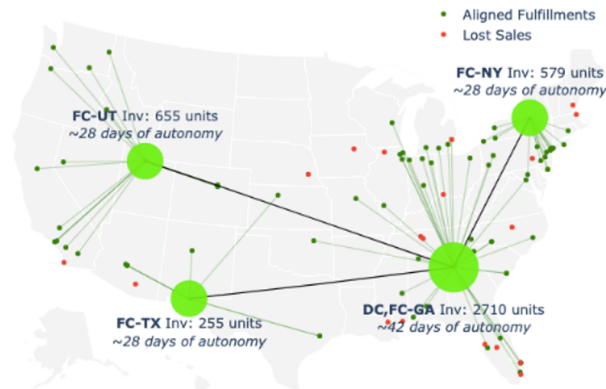


Figure 2: Inventory level and flow of goods through the network

- Advanced Analytics and Visualizations:** Utilize the simulator's analytical tools to monitor key performance indicators, such as market penetration, demand fulfillment, and profit indicators, displayed on the top bar. Further, many plots are included to enable you to visualize the impact of your decisions on the various aspects of the complex network.

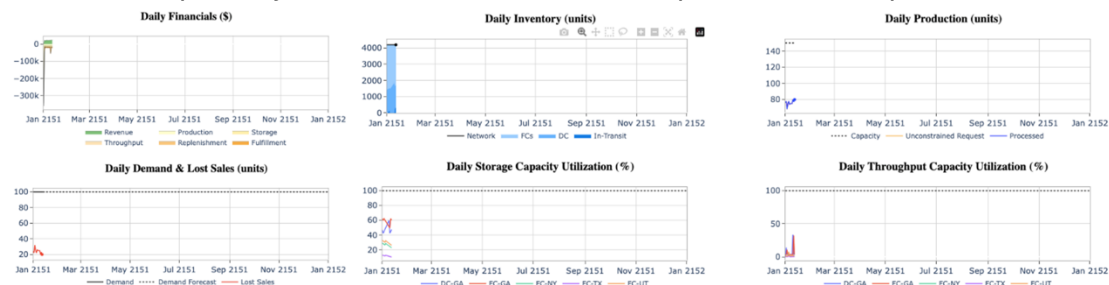


Figure 3: Visualizations embedded in the simulator

- Scenario Testing:** Experiment with different demand scenarios and transportation modes to observe their impact on customer satisfaction and cost efficiency. The simulator allows you to adjust parameters to find the optimal balance between the various tradeoffs, and manage the intricacies of this complexity with a thoroughly researched analytical background.

Overall, we have developed the simulator for you – to enable you to perceive, understand, and appropriately react to the various aspects in a real-world-inspired supply chain. We envision this to be an aid for you, as you prepare for a journey towards becoming supply chain leaders of the future.

*Note: The Tsukumo company at the core of this casework does not exist.
All stated facts are fictitious, and all data has been synthetically generated.*

Overall Case Introduction

The Tsukumo Company is an established yet small player in the design, engineering, manufacturing, commercialization, and distribution of domestic collaborative robots. It has an excellent portfolio of products with four families ranging from smaller to larger capability and capacity. Tsukumo has decided early on to be fully e-commerce based. It has grown slightly in the last 10 years to reach a current 3.6% market share of an annual estimated continental-USA market of 2,000,000 product units for its types of collaborative robots.

Originally, Tsukumo manufactured its products in Japan, where it achieved tremendous success in the domestic market. Building on this success, it entered the U.S. market, but as volumes have grown, the company has struggled to properly align supply and demand due to long international lead times, rising uncertainties, and increasing tariff-related costs on imports. To address these challenges, Tsukumo is now exploring a transition to U.S.-based manufacturing, paired with a network of strategically located fulfillment centers, to better balance responsiveness, efficiency, and cost competitiveness.

Tsukumo has revitalized its core leadership team in the last three years. Tsukumo's lead team senses that the company could reach much higher market shares and attain operational excellence with a combination of (1) improved business leadership, (2) better alignment of demand and supply throughout its network through coordination and collaboration, (3) transitioning from owned facilities to a flexible network embedding third-party logistics partners, and (4) leveraging data-driven daily forecasting and AI-based decisional models.

The lead team has carefully studied the overall market. It has interviewed facility managers, current and potential customers, as well as third-party storage and logistics providers and experts from the industry. It has also purchased market databases and analyzed data relative to multiple key market drivers. Combining all this, it has made a set of discoveries and hypotheses.

Your role as a supply chain consultant team contracted by Tsukumo is to help analytically model, validate, and visualize the hypotheses through tasks and simulations to enable the leadership to take the right actions about its network of supply chain facilities, towards overall operational excellence and growth.

Your work helping Tsukumo throughout this casework is structured in two major phases. First is the preparatory analysis and planning phase 1, performed without access to the interactive simulator. This phase focuses on systematically tackling Tsukumo's five key sets of discoveries and hypotheses respectively relative to market demand, fulfillment network, delivery time sensitivity, robust autonomy targeting, and network replenishment.

Upon phase 1 completion, second is the simulation-based holistic analysis and planning phase 2. This phase integrates tackling all Tsukumo's discoveries and hypotheses in light of the learnings from phase 1, leveraging an iterative multi-level gamified approach.

Phase 1: Preparatory analysis and planning

Discovery and Hypothesis Set 1: Market Demand

- The overall market in the USA is expected to grow by about 7.5% over the next year, with conservative and optimistic scenarios at 4% and 12% yearly. Unless a task states otherwise, apply the growth rate to the current 2,000,000-unit annual market to obtain the market demand over the planning horizon.
- Given its new strategic orientation with strong presence within the USA, Tsukumo's market share is targeted to grow yearly by a minimum of 15%, a median of 20%, and a maximum of 25% from its current 3.6% market share.
- The overall market has been categorized into three market types. Tsukumo has identified that its primary markets are 100 miles radius around the center of key Metropolitan Statistical Areas (MSAs) across the USA, as in *msa.csv*; secondary market as 300 miles radius around these MSAs, beyond the primary markets; and tertiary markets as all other US territories. The corresponding market type for each 3-digit ZIP is provided in *zip3_market.csv*. (Figure 4)

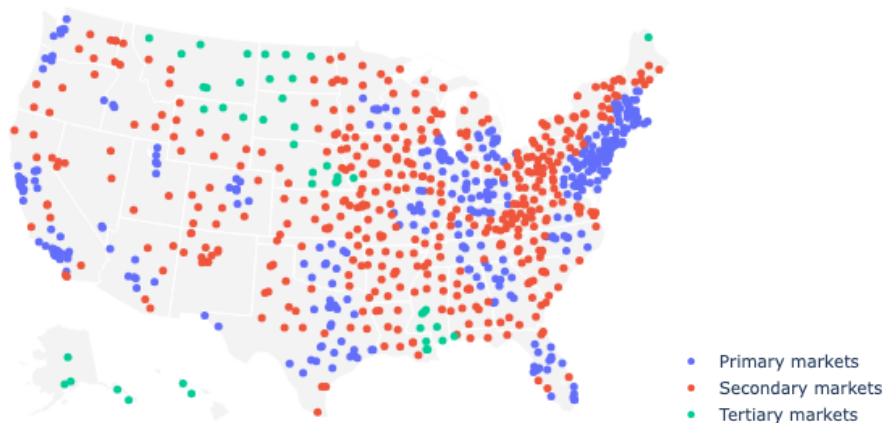


Figure 4: Market Type Map (points represent 3-digit ZIPs)

- Based on historical information and market analysis, demand for Tsukumo products is subject to seasonal effects. Expectations of week-of-year and day-of-week seasonalities are provided in *demand_seasonalities.csv*. Respectively assume 20% and 15% coefficient of variation on these estimates.
- The expected probability mass function (PMF) for geomarket demand share at each 3-digit ZIP in USA can be approximated to remain stable over the next five years, as provided in *zip3_pmf.csv*. Assume a 15% coefficient of variation on these estimates. The shares provided sum to one. A small number of 3-digit ZIPs appear in *zip3_market.csv* without a row in *zip3_pmf.csv*; treat their expected demand share as zero.
- The average price (\$), weight (lbs) and volume (width*height*length ft³) of packaged Tsukumo products are to be on the order of \$3,000, 60lbs, and 3*2*2 ft³.

- Throughout this casework, express your demand estimates using the notation seen in class: d_{pmt} for the market demand of product family p in market m over period t , covering Tsukumo and all its competitors, and d_{cpmt}^s for the share of that demand captured by company c , so that Tsukumo's own demand is d_{cpmt}^s with $c = \text{Tsukumo}$. Index t spans the historical periods H over which you fit seasonality and uncertainty, and the future periods F of the planning horizon. The data provided aggregates the four product families, so index p may be dropped unless you choose to model the families separately.

Task 1

Assuming no demand uncertainty and no demand seasonality, compute the estimated overall market demand (for Tsukumo and its competitors) over a 1-year planning horizon for the various combinations of the three aggregation threads below:

- Temporal: (Y) Yearly, (M) monthly, (D) daily*
- Spatial: (A) USA territory, (B) Market Type, (C) State, and (D) 3-digit ZIP*
- Measure: (\$) dollars, (#) units, (V) volume, (W) weight*

An example of combination is D-B-\$, corresponding to daily demand in \$ segregated by market type. Present the combinations you judge most informative, with visual displays, and draw actionable insights from them.

Task 2

Repeat task 1 now accounting for demand seasonality and uncertainty. Instead of deterministic numbers as in task 1, the demand estimates take the shape of statistical distributions and their $X\%$ -min, mode, mean, $X\%$ -max values. Consider 68% (1σ), 95% (1.65σ), and 99% (2.33σ) for X .

Model the executive growth targets as triangular distributions, using the stated minimum, mode, and maximum for both the overall market growth (4%, 7.5%, 12%) and Tsukumo's market share growth (15%, 20%, 25%). Generate a set of about 25 demand scenarios by inverse transform sampling from these triangular distributions, as done in class, and propagate each scenario through the week-of-year and day-of-week seasonality factors and the zip3 demand share PMF, with their respective coefficients of variation. Report the r -robust demand bounds $\bar{d}_{pmt} = d_{pmt} + z_r \sigma_{pm}$ for each robustness level considered, and state which sources of variability each reported bound accounts for.

Discovery and Hypothesis Set 2: Fulfillment Network

- Facilities are identified by their type, state, and 3-digit zip code.
 - In the facility typology used throughout the course, AF-GA-303 is a production fab, DC-GA-303 and the fulfillment centers are deployment centers, and the network of logistic hubs operated by the third-party logistics service provider introduced in Discovery and Hypothesis Set 5 supports the transit and consolidation of replenishment shipments.
 - Tsukumo is completing implementation of a single assembly factory (AF-GA-303) in the USA, located in Georgia within zip code 303.
 - Tsukumo has acquired a single US-based distribution center (DC-GA-303) also located in Georgia's zip code 303, fed by nearby AF-GA-303 and feeding all its new fulfillment centers (FCs) within its US-based fulfillment network.
 - Tsukumo is considering three alternative fulfillment networks:
 - Single-FC Network, centralizing all fulfillment operations at FC-GA-303, fed by its nearby distribution center.
 - Four-FC Network, relying on FCs in Georgia (FC-GA-303), New York (FC-NY-134), Texas (FC-TX-799), and Utah (FC-UT-841), as displayed in Figure 5.
 - 15-FC Network, relying on the following FCs, as displayed in Figure 6:
- | | | | | |
|--------------|--------------|--------------|---------------|---------------|
| 1. FC-AZ-852 | 4. FC-CO-802 | 7. FC-IL-606 | 10. FC-NC-275 | 13. FC-TX-770 |
| 2. FC-CA-900 | 5. FC-FL-331 | 8. FC-MA-021 | 11. FC-NJ-070 | 14. FC-UT-841 |
| 3. FC-CA-945 | 6. FC-GA-303 | 9. FC-MI-481 | 12. FC-TX-750 | 15. FC-WA-980 |

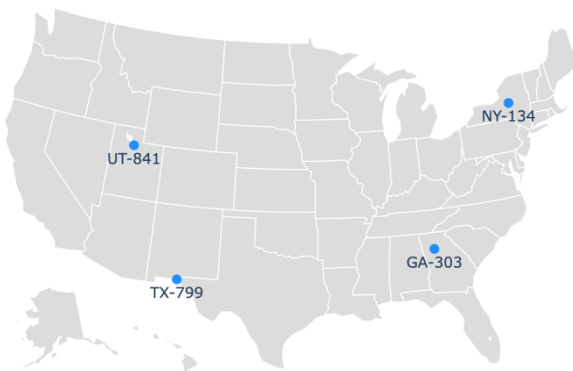


Figure 5: 4-FC Fulfillment Network

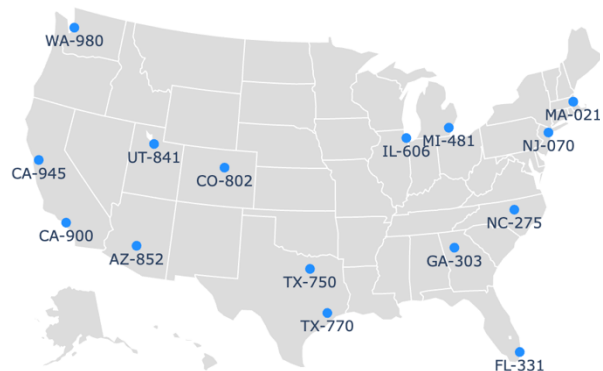


Figure 6: 15-FC Fulfillment Network

Task 3

Based on location proximity, estimate target regions for each FC, where all the 3-digit ZIP codes are strictly served by the closest FC. This yields clusters of 3-digit ZIPs with the same preferred FC.

- Provide a geographical plot coloring the various 3-digit-zipcode clusters.
- Estimate the aggregated demand share for each FC, and for each FC and market type combination.

- c) *Estimate the demand distribution for each market type by assigning shipments into distance buckets based on the mileage from each FC to its served 3-digit ZIPs. Use the following shipping distance buckets (in miles):*

≤ 50 , 51-150, 151-300, 301-600, 601-1000, 1001-1400, 1401-1800, >1800

Distance from FC locations to all 3-digit ZIPs is provided in `fc_zip3_distance.csv`.

Task 4

Tsukumo's lead team is keen on ensuring resilience through multi-source fulfillment, so it allows to ship customer orders from a farther FC than the preferred one to avoid losing a sale when the closest FC is stocked out or disrupted. With a goal to minimize lost sales, identify for each 3-digit ZIP the set of FCs that can serve as fulfillment centers within the same distance bucket as the closest FC, or in the next higher bucket of the list given in task 3-c. This enables creating fulfillment clusters, each including the set of 3-digit ZIP codes with the same set of FCs.

- a. *Provide a geographical plot coloring the generated fulfillment clusters.*
- b. *Provide a geographical plot coloring the various clusters based on the number of FCs that can serve the cluster, in a red (lowest) – yellow – green (highest) color scheme.*
- c. *Estimate the proportion of demand that can only be served by a single FC.*
- d. *Compute the demand distribution across the shipping distance buckets (as defined in task 3-c), under the following fulfillment rule: (i) For each 3-digit ZIP, allocate 80% of its demand to the closest FC; (ii) Allocate the remaining 20% of demand equally among all other FCs that are part of the fulfillment cluster.*

Discovery and Hypothesis Set 3: Delivery Time Sensitivity

- In this era of shrinking order-to-delivery (OTD) time promises by e-commerce retailers and induced e-customer expectations, the proportion of customers willing to place orders with Tsukumo depends on promised OTD times and this proportion differs per market type. Table 1 shows for example that in all primary markets, it is expected that only 60% of the demand can be converted into sales if the promised OTD time is four days. In Tables 1 and 2, a column headed by a number of days denotes a promise to deliver within at most that many days, and the 5+ column denotes a promise of more than 5 days.

Market	Promised Order-To-Delivery Lead Time (days)					
	1	2	3	4	5	5+
Primary	100%	90%	75%	60%	40%	30%
Secondary	100%	100%	95%	75%	60%	40%
Tertiary	100%	100%	100%	95%	80%	60%

Table 1: OTD Time and Market Type Sensitive Expected Demand Conversion Rates

- All markets must be covered through white-glove online-based sales experience with corporate-led fulfillment and service, while delivery is to be leveraging third-party delivery service providers (DSPs). The product shipment costs per unit from FCs to customer location depends on the promised OTD time and shipment distance zones, as denoted in Table 2. Tsukumo sells and delivers exclusively to the customer location. Given the value of the products and the white-glove service accompanying them, it offers no store, click-and-collect, or locker pickup alternative, so its network includes no customer interface facilities. Table 2 prices every promise in every distance zone, so any OTD promise can be offered in any zone. Choosing a promise is therefore an economic decision, not a feasibility one.

Zone (miles)	Promised Order-To-Delivery Lead Time (days)					
	1	2	3	4	5	5+
1 (≤ 50)	\$607	\$353	\$230	\$139	\$121	\$103
2 (51-150)	\$759	\$441	\$287	\$173	\$151	\$128
3 (151-300)	\$1025	\$585	\$392	\$238	\$191	\$160
4 (301-600)	\$1445	\$794	\$533	\$316	\$242	\$198
5 (601-1000)	\$2078	\$924	\$655	\$343	\$287	\$225
6 (1001-1400)	\$2692	\$1427	\$895	\$491	\$340	\$259
7 (1401-1800)	\$2841	\$1795	\$1202	\$776	\$362	\$276
8 (> 1800)	\$2912	\$1854	\$1330	\$894	\$388	\$301

Table 2: Shipment Costs (per product unit)

Task 5

Using the closest-FC assignment of task 3, consider each market type and each OTD time promise option from 1 to 5+ days. The demand conversion rate follows from Table 1, given the market type and the promise. The shipment cost per unit follows from Table 2, given the promise and the distance zone between each 3-digit ZIP and the FC assigned to it in task 3. For each market type and each

option, estimate the induced (a) achievable total revenue assuming that the average selling price of a product unit is \$3,000, (b) total shipping cost, (c) net revenue (revenue – shipping cost), and (d) gross operating profit, defined as net revenue minus cost of goods sold of \$750 per unit covering raw materials and production. This profit measure excludes AF, DC, and FC capacity costs and DC and FC replenishment costs, which are only estimated in later tasks. Plot and discuss your results.

Task 6

Propose an optimized OTD time promise for each market type, with aim to maximize expected overall gross operating profit. Document and justify your optimization model or heuristic algorithm. Compute the resulting expected revenue, shipping cost, net revenue, and gross operating profit in each market type and over the overall market. Contrast with the two extreme solutions corresponding to offering 1-day service to all and offering 5+-day service to all.

Discovery and Hypothesis Set 4: Robust Autonomy Targeting

- Tsukumo aims for each FC to maintain a 3-week-99% target autonomy stock. This signifies that it aims to have at least enough inventory at all times to robustly cover without replenishment the upper 99% forecasted demand over the next 3 weeks, aiming not to have more stock than required by this target. This autonomy target, expressed in time (weeks) at a given robustness (%), implies a target inventory that dynamically evolves on a daily basis depending on expected seasonality, growth, and uncertainty.
- Tsukumo aims to maintain a 6-week-99% target autonomy stock in its overall fulfillment and distribution network (FCs+DC), adjusted dynamically on a daily basis. Given the above bullet, the complementary inventory is intended to be held at the DC.

Task 7

Estimate the stock to be maintained in each FC in the fulfillment network for each day in the planning horizon given the 3-week-99% autonomy currently targeted by Tsukumo's lead team, building on the results from the previous tasks. Compute the estimated maximum inventory over the planning horizon induced by the target autonomy policy at each FC and over the fulfillment network. Compute the robust demand to be covered over the autonomy period with the expression seen in class, $D_{ls} = d_b (l/b) + z_s \sqrt{(\sigma_b^2 l/b)}$, where l is the autonomy period, b the base period over which your demand statistics are estimated, d_b and σ_b the mean and standard deviation of demand per base period, and z_s the robustness factor associated with the targeted service level.

Estimate the stock to be maintained in the overall fulfillment and distribution network for each day in the planning horizon given the 6-week-99% target autonomy currently targeted by Tsukumo's lead team. Accounting for the stock expected to be maintained in each of the FCs according to their autonomy policy, estimate the stock to be maintained in the DC for each day in the planning horizon to ensure there is enough inventory in the network to respect the 6-week-99% target autonomy.

Contrast these results with those induced by combinations of alternative overall fulfillment and distribution network autonomy periods of 4 and 8 weeks and of robustness of 50%, 68%, and 95%. Present the contrast in the paired table format used in class, with the autonomy period down the rows (4, 6, and 8 weeks) and the robustness level across the columns (50%, 68%, 95%, and 99%), reporting for each combination the maximum inventory induced at the FCs, at the DC, and over the network. Discuss your results.

Task 8

Given the results of task 7, estimate the daily production capacity requirements at the assembly factory if it is to feed the DC with additional products daily to ensure that it reaches the stock level enabling respect of the 6-week-99% autonomy target. Estimate the maximal daily production capacity required over the planning horizon. This corresponds to a pursuit production strategy, where production tracks the evolving requirement day by day.

Consider next a full smoothing strategy without pre-production, where the AF holds a constant production rate from the first day of the planning horizon. Determine whether that rate is capable of covering demand throughout the year, and in particular through the seasonal peaks. Report the magnitude and duration of any backlog it creates, and the DC inventory it accumulates.

As an alternative, consider a full smoothing strategy with pre-production, where Tsukumo holds a steady production rate at the AF but starts producing ahead of the peaks rather than on the first day of the horizon. This will cause Tsukumo to inject in the DC inventory beyond what is required by the robust 6-week-99% network-wide autonomy target policy at some times during the year, so as to absorb the smoothing-induced anticipatory production. Report the pre-production capacity and the accumulated inventory this strategy requires to become capable.

Consider finally a segmented smoothing strategy, where the planning horizon is partitioned into a small number of segments, the production rate being held constant within each segment while allowed to change from one segment to the next. Propose and justify your segmentation against the seasonality profile, and estimate the induced AF production capacity requirement and the DC inventory carried over the horizon.

Compare the three strategies on the maximum daily AF production capacity required, the maximum DC inventory carried, and the profile of the production requirement over the planning horizon, presenting the contrast in the format used in class. State which strategy you retain and why.

Discovery and Hypothesis Set 5: Network Replenishment

- Tsukumo is considering utilizing a third-party logistics service provider (LSP) to manage replenishment of the fulfillment network. This enables Tsukumo to not be constrained by its fleet of trucks, rather utilizing the LSP's fleet and network and gain consolidation benefits. Tsukumo and the LSP have reached an agreement on the shipment costs, with the conditions that (1) shipments must be done using tractors pulling a 53-foot semi-trailer capable of holding 272 product units, corresponding to a full-truck-load (FTL) and (2) no shipment must depart from the DC with the semi-trailer containing less than 10% of a FTL. The replenishment shipping costs per unit are provided in Table 3.

Zone (miles)	Replenishment Shipment Cost	Zone (miles)	Replenishment Shipment Cost
1 (≤ 50)	\$48.3	5 (601 - 1000)	\$114.9
2 (51 - 150)	\$60.4	6 (1001 - 1400)	\$136.0
3 (151 - 300)	\$76.2	7 (1401 - 1800)	\$144.7
4 (301 - 600)	\$96.6	8 (> 1800)	\$155.2

Table 3: FC Replenishment Shipment Cost (per product unit)

- The LSP, through its network of logistic hubs, is able to swap drivers and/or semi-trailers to achieve faster shipment durations, respecting the daily driving limit for truckers in the US. For simplicity, consider that shipments progress at a velocity of 500 miles/day, accounting for additional hours for consolidation and distribution. For your ease, Table 4 provides the time at which replenishments have arrived and are available to fulfill customer demand.

FC ID	RAD	FC ID	RAD	FC ID	RAD	FC ID	RAD
FC-GA-303	T	FC-AZ-852	T+4	FC-IL-606	T+2	FC-NJ-070	T+2
FC-NY-134	T+2	FC-CA-900	T+4	FC-MA-021	T+2	FC-TX-750	T+2
FC-TX-799	T+3	FC-CA-945	T+5	FC-MI-481	T+2	FC-TX-770	T+2
FC-UT-841	T+4	FC-CO-802	T+3	FC-NC-275	T+1	FC-WA-980	T+5
		FC-FL-331	T+2				

Table 4: FC Replenishment Availability Day (RAD) when it departs from DC-GA-303 on day T

- To respect the 3-week 99%-robust FC autonomy targets aimed to manage the balance between the risk of stock-out situations and the wasteful cost of overstock situations, Tsukumo's supply chain team proposes an FC replenishment algorithm using a combination of replenishment interval and minimum robust autonomy threshold, in line with the previous hypotheses and tasks. Its essence is hereafter described through an example.

Consider an FC with a 7-day replenishment interval and a 14-day 99%-robust minimum autonomy threshold. The algorithm tracks the FC's current robust autonomy daily, i.e. the number of days till the FC will stock-out given its inventory position (on-hand inventory plus replenishments enroute, if any) and the upper 99% demand forecast for the forthcoming days, assuming no additional replenishments are provided. When the robust autonomy days

fall below 14 days, a replenishment is triggered. The replenishment quantity is the greater of (3-week 99%-robust autonomy target inventory - inventory position) and 10% FTL (27 units).

If the 14-day minimum robust autonomy days threshold is not breached on any day within the 7-day replenishment interval, a replenishment decision is triggered on the 7th day from last shipment. In this case, the replenishment quantity is again determined to bring the inventory position to the 3-week 99%-robust autonomy target, respecting the 10% FTL threshold. If the FC does not require replenishment (current 99%-robust autonomy $\geq$ 21 days), then order is not placed. The algorithm continues similarly for subsequent intervals.

Task 9

Based on the solution of your tasks so far, and considering a 7-day replenishment interval and 14-day minimum robust autonomy threshold, estimate the profile of daily replenishment quantities over the planning horizon for each FC.

Then, estimate the throughput capacity required daily for each FC and the DC. The daily throughput capacity is defined as the maximum capacity to handle both inbound and outbound movement in a single day. Estimate the throughput capacity required at each FC and DC to robustly support replenishments and customer order fulfillment, in product units, over a 1-year planning horizon. For the inbound DC throughput capacity, contrast the requirements across the three production strategies of task 8.

Then convert these unit throughput requirements into resource requirements using the capacity relation seen in class, $C = d t / (q a r e)$, where d is the number of units to be handled over the period, t the unit handling time, q the quality yield, a the availability, r the reliability, and e the efficiency of a handling resource. Assume a unit handling time of 12 minutes, a quality yield of 0.98, an availability of 0.90, a reliability of 0.95, an efficiency of 0.90, and a single 8-hour shift per day. Compute the resulting number N of handling resources required at each FC and at the DC, applying the decision rule seen in class for a fractional capacity requirement with a threshold f of 0.3, and justify how any residual load is absorbed when you round down. Conduct a sensitivity analysis on q , r , and e , reporting how the required N and the induced capacity cost respond.

Finally, decompose the storage capacity requirements obtained in tasks 7 and 8 into the three contracting tiers seen in class: base capacity, held year-round in internal or long-term contracted space; seasonal capacity, leased under contract for a committed block of consecutive weeks; and peak-and-extreme capacity, acquired on demand on the spot market. Using the storage rates in Appendix 1, propose for each FC and for the DC a tier split covering the daily inventory profile at the lowest annual storage cost, and report the volume and the cost carried by each tier.

Task 10

The 7-day interval and 14-day threshold used in Task 9 were assumed uniformly across all FCs. Propose instead a replenishment interval and minimum robust autonomy threshold for each FC individually, and justify them.

Two considerations should drive your choice. First, the threshold must leave enough robust autonomy to cover the FC's replenishment shipment duration (Table 4) plus a margin for forecast

error accumulated over that window. Second, the interval should be long enough that the FC's typical order quantity comfortably exceeds the 10% FTL minimum of 27 units, so that replenishments are not routinely inflated above what the autonomy target requires. Note that inflated replenishments push inventory above the 3-week-99% target and incur storage cost accordingly.

- a) Derive the relationship between the interval and the threshold implied by the 3-week autonomy target, and explain which of the two triggers governs under your chosen parameters.*
- b) Propose parameters for each FC. Present them in a table alongside the FC's demand share, shipment duration, and implied average inventory.*
- c) Report, for your proposed parameters, the frequency with which replenishments are floored at 27 units and the resulting average inventory, and contrast both against the uniform 7/14 configuration from Task 9.*
- d) Identify any FC whose demand volume is low enough that the 10% FTL minimum cannot be respected without systematically exceeding the autonomy target. Discuss what this implies for the viability of that FC in the network.*

Submission of tasks 1 to 10 deliverables enables access to the simulator and documentation necessary to engage in Phase 2.

Appendix 1. Financial information

- *Revenue - \$3,000 for each unit of product demand fulfilled*
- *Production*
 - *Cost of raw materials: \$600 per unit produced*
 - *Cost of production equipment utilization: \$150 per unit produced*
- *Production Capacity*
 - *Operation and maintenance cost of contracted capacity: \$100 per unit per day*
 - *One-time capacity setup or increase - \$1,400 per unit increase, \$0 when decreased*
- *Storage Capacity*
 - *Base capacity, contracted year-round: operation and maintenance cost of \$6.60 per unit per day*
 - *One-time base capacity setup or increase - \$46.20 per unit increase, \$0 when decreased*
- *Throughput Capacity*
 - *Operational cost of handling throughput movement - \$25 per unit of utilized capacity*
 - *Operation and maintenance cost of contracted capacity: \$3.60 per unit per day*
 - *One-time capacity setup or increase - \$25.20 per unit increase, \$0 when decreased*